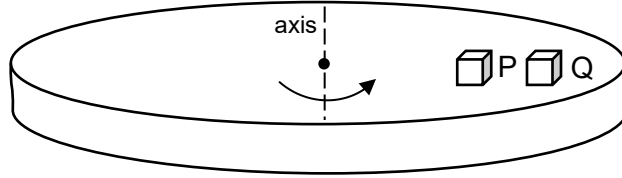


- 9** Two identical objects, P and Q, are placed on a flat rough circular disc. Q is at a further distance from the centre of the disc compared to P.



The disc starts spinning from rest about its central axis with increasing speed. At a certain speed, one of the objects starts to slide off the disc.

Q slides off the disc first because

- A** the friction experienced by P and Q are always equal.
- B** it moves with a larger angular velocity compared to P.
- C** it requires greater friction to keep it in its position compared to P.
- D** the friction experienced by P is larger than Q before one of the objects slide off the disc.

Answer: C

$$F_c = mr\omega^2$$

Both objects have the same mass and same angular velocity, but the centripetal force required for Q's circular motion is larger due to larger radius.

When the centripetal force required exceeds the maximum static friction available, Q starts to slide.

10 P and Q are two points above Earth's surface at distances r and $2r$ respectively from the